

Motor Control Board

Variant: DRAFT

2026-08-17

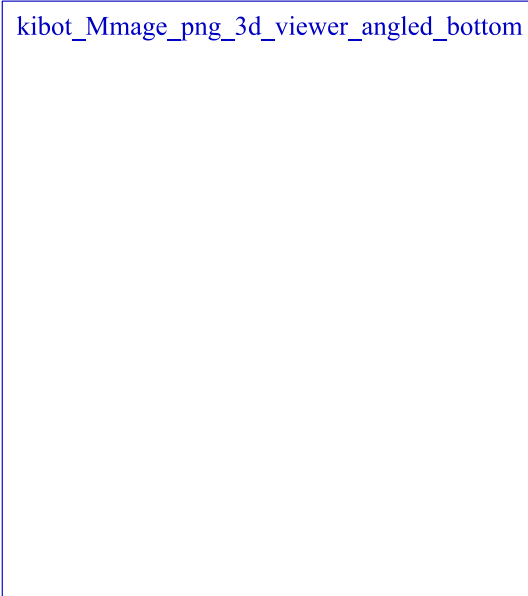
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

NOTES

Comment

Not fitted components are marked as **X**

- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Date: 17-Aug-2026

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

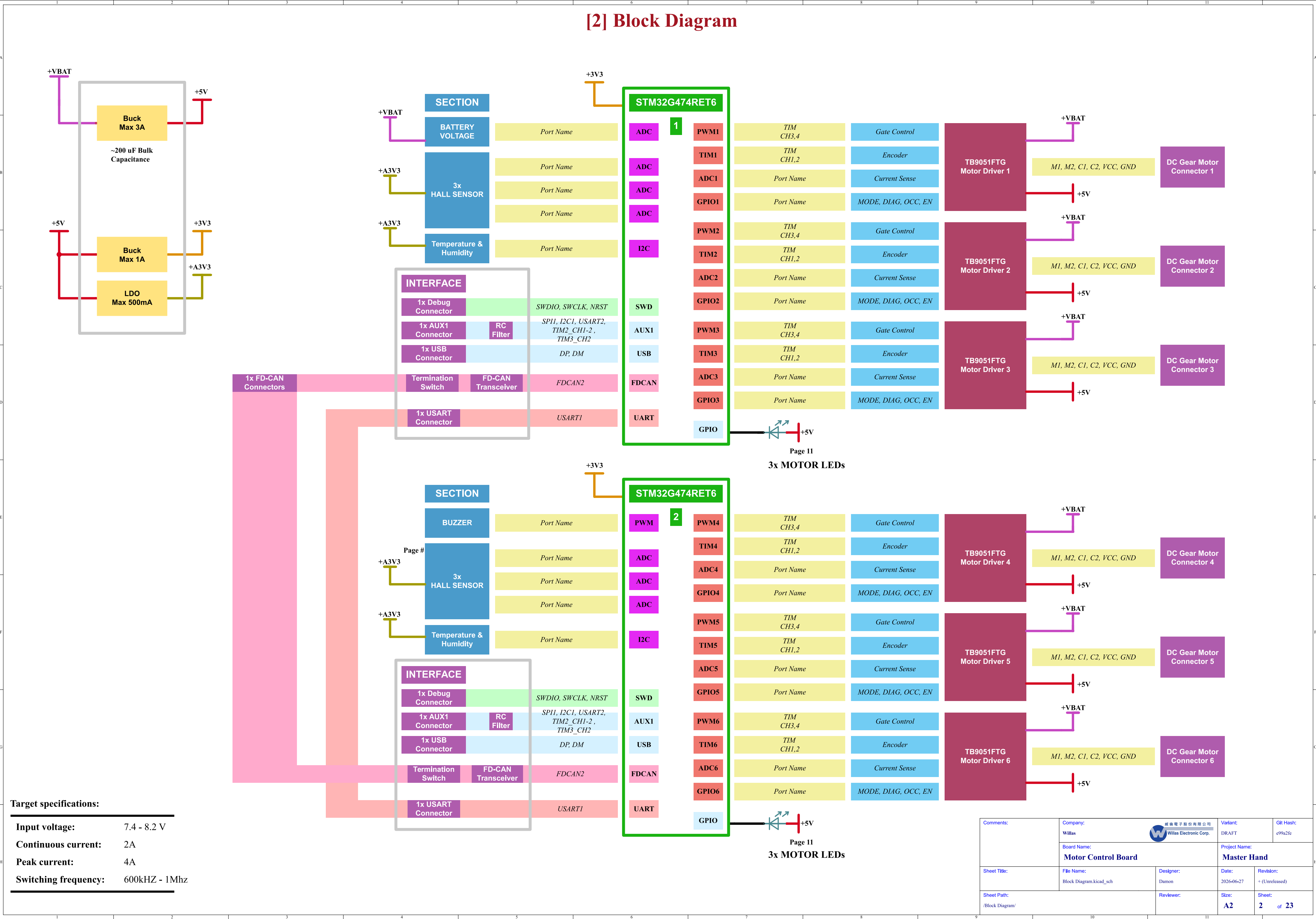
File: Project Architecture.kicad_sch

Power - Sequencing

File: Power - Sequencing.kicad_sch

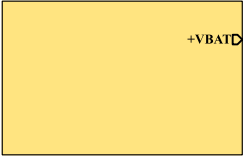
Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fc
	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /			Reviewer:	Size: A3	Sheet: 1 of 23	

[2] Block Diagram

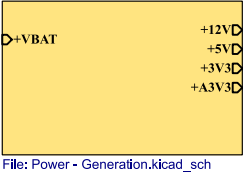


[3] Project Architecture

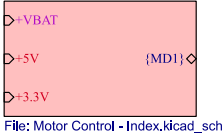
Power - Connectors



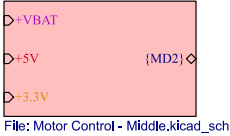
Power - Generation



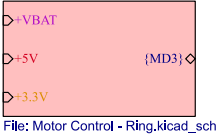
Motor Control - Index



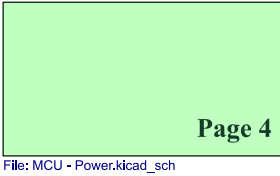
Motor Control - Middle



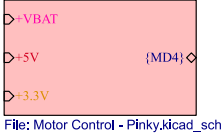
Motor Control - Ring



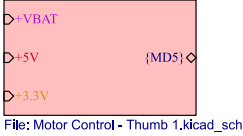
MCU - Power



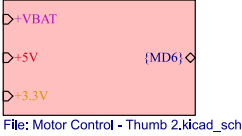
Motor Control - Pinky



Motor Control - Thumb 1



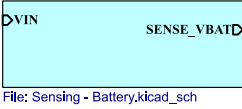
Motor Control - Thumb 2



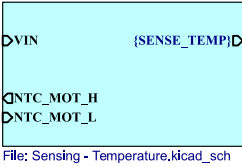
MCU - IOs



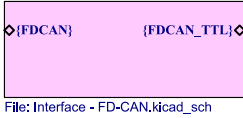
Sensing - Battery



Sensing - Temperature & Humidity



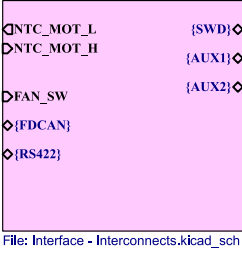
Interface - FD - CAN



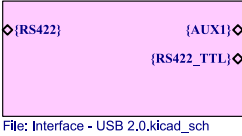
Misc - Holes, Fiducials



Interface - Interconnects



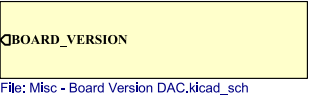
Interface - USB 2.0



User - LED Indicators & Buzzer

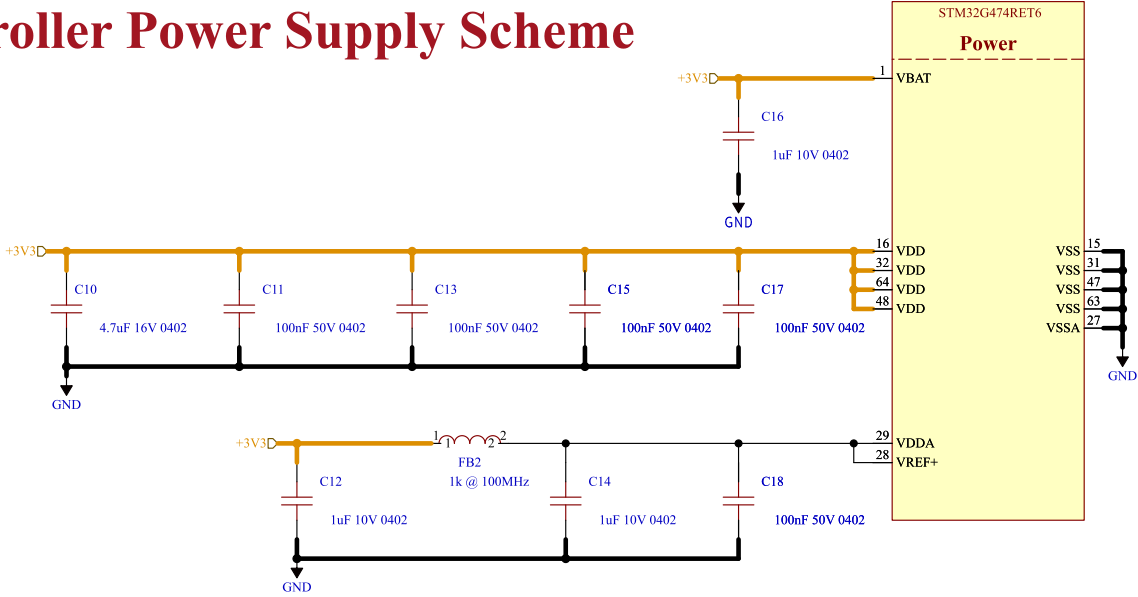
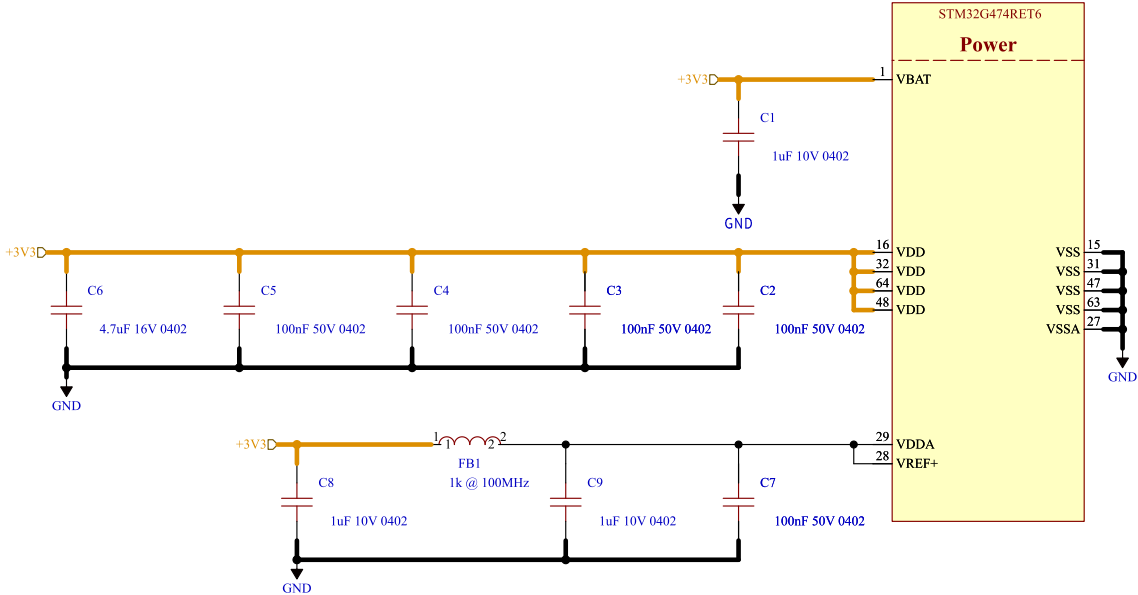


Misc - Board Version, DAC



Comments:	Company: Willas 		Variant: DRAFT	Git Hash: e99a2fc
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Project Architecture.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 23

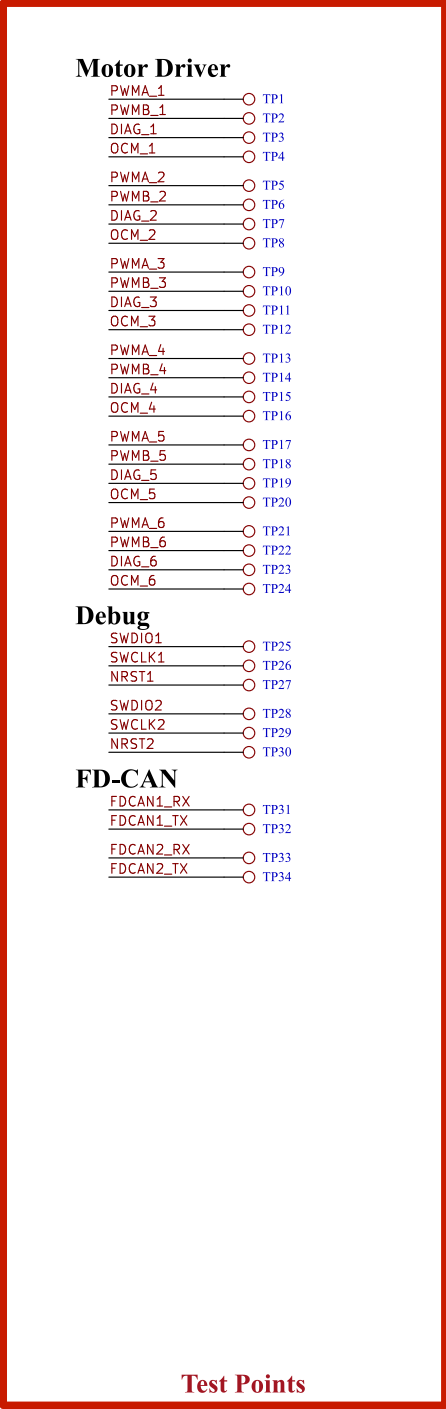
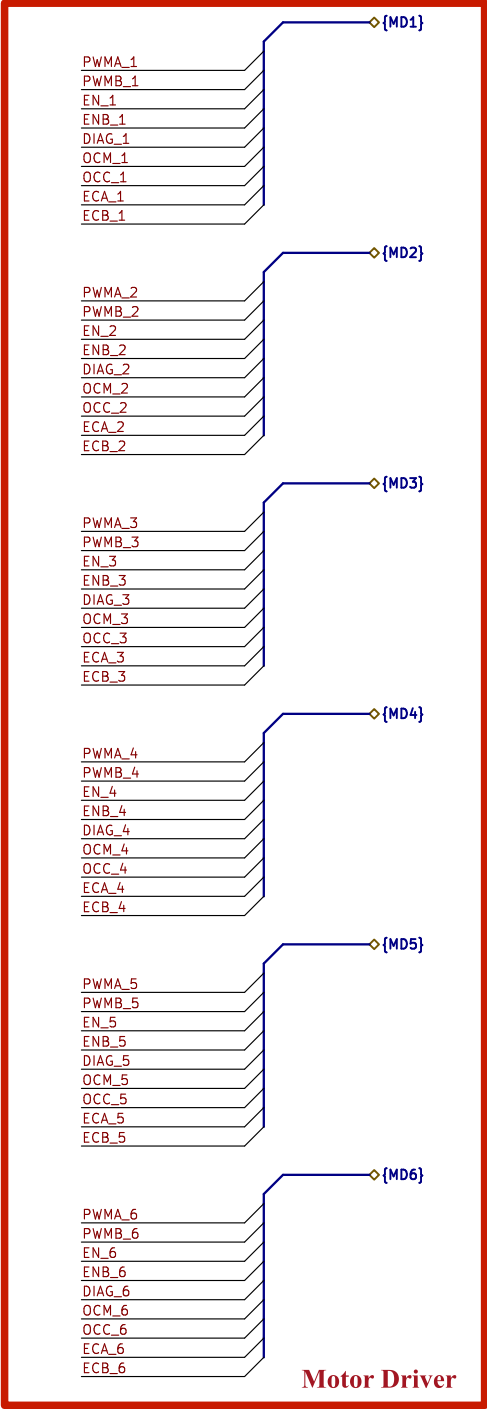
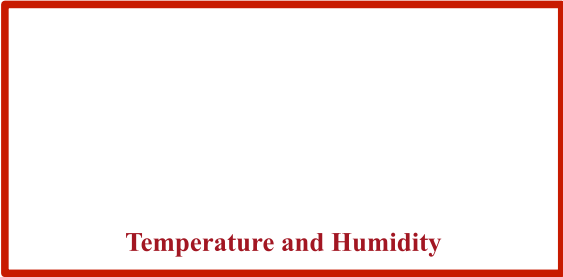
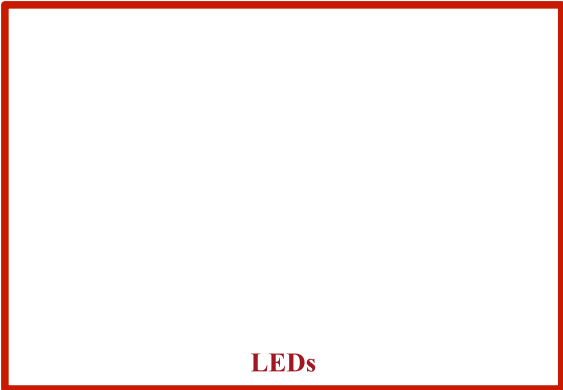
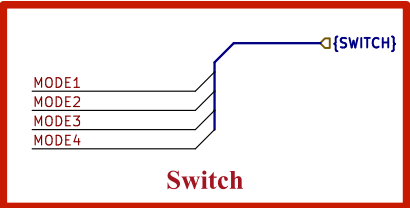
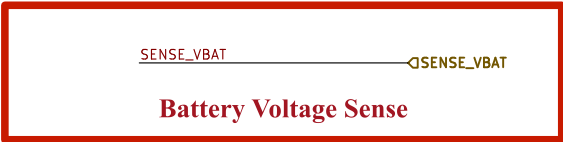
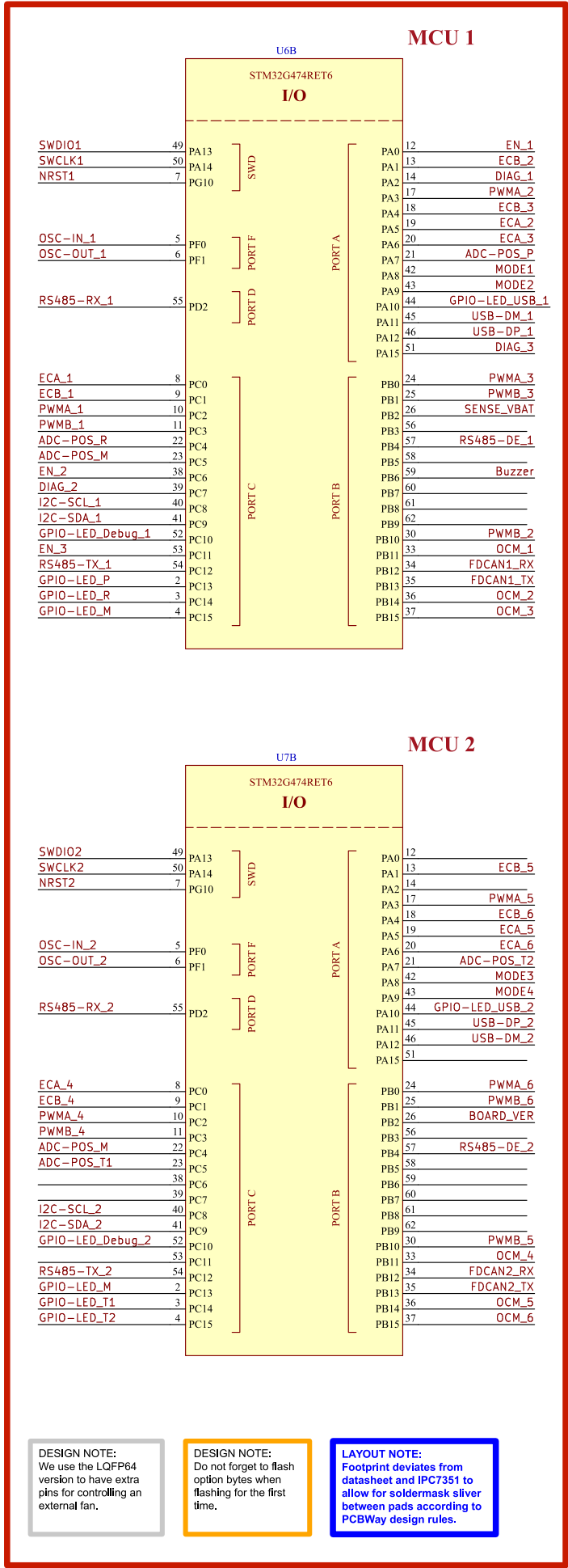
Microcontroller Power Supply Scheme



LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.

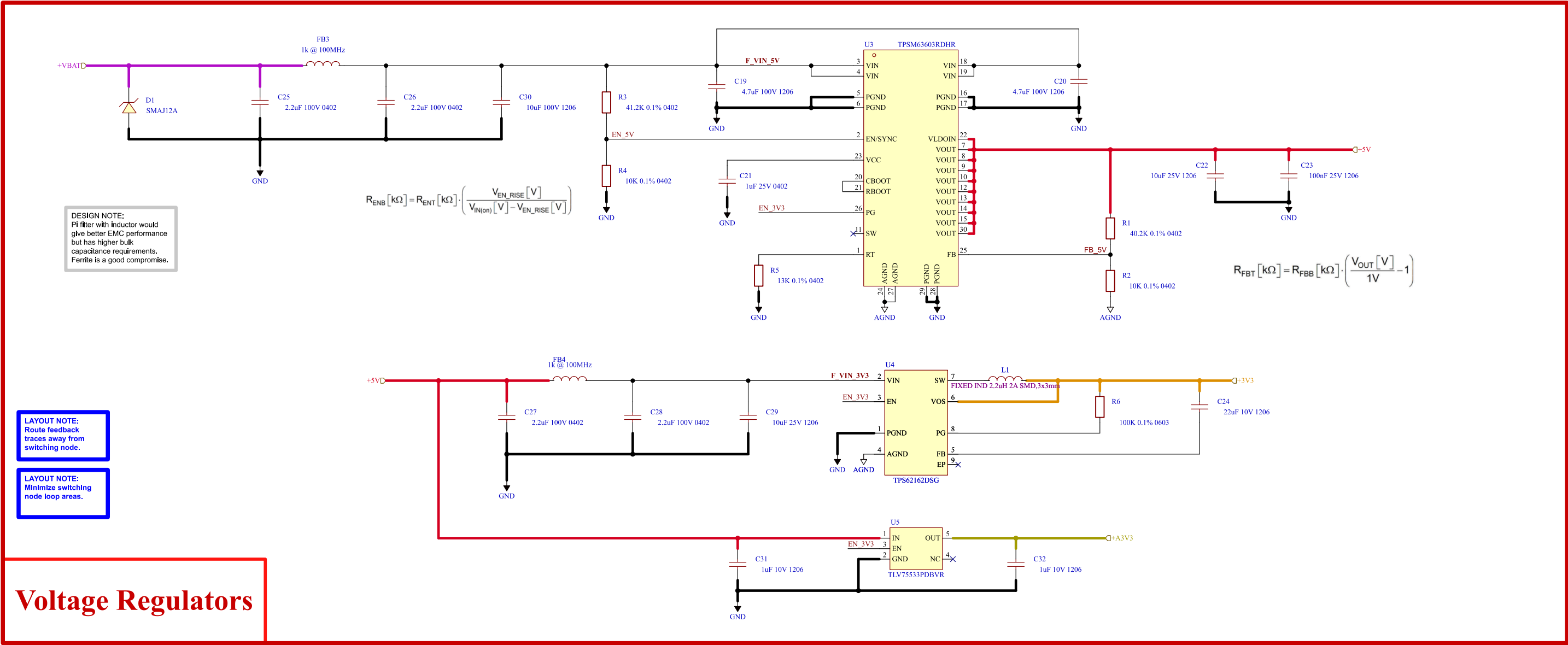
Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fc
	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: MCU - Power.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/MCU - Power/			Reviewer:	Size: A3	Sheet: 4 of 23	

[5] MCU - IOs

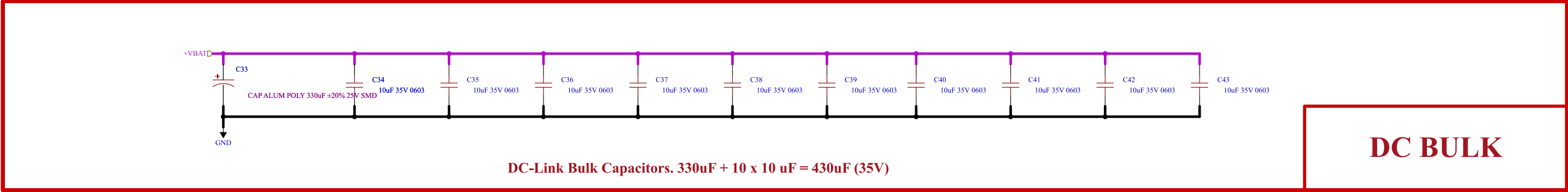


Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fc
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Sheet Title:	File Name: MCU - IOs.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
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[6] Power - Generation



Voltage Regulators

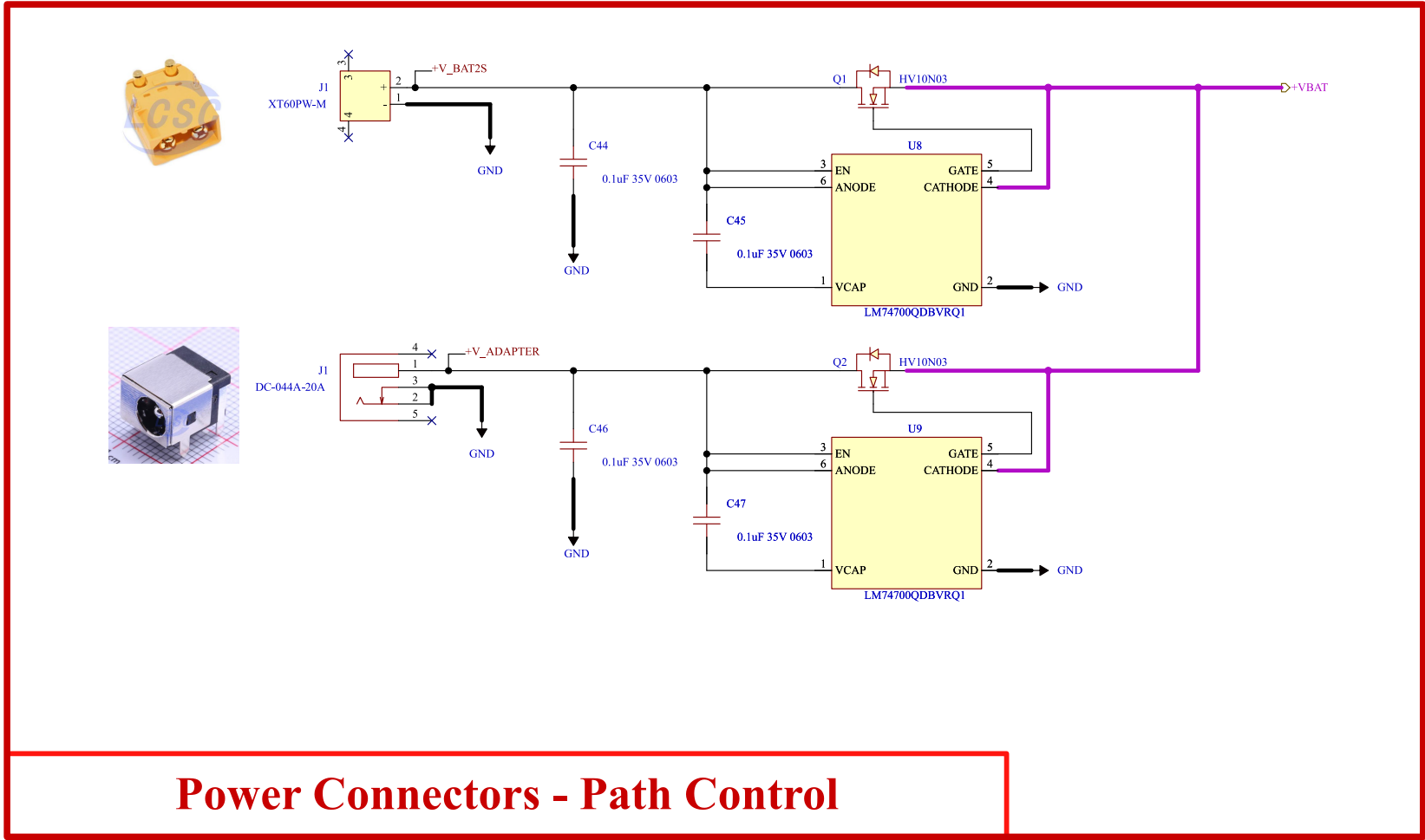


DC BULK



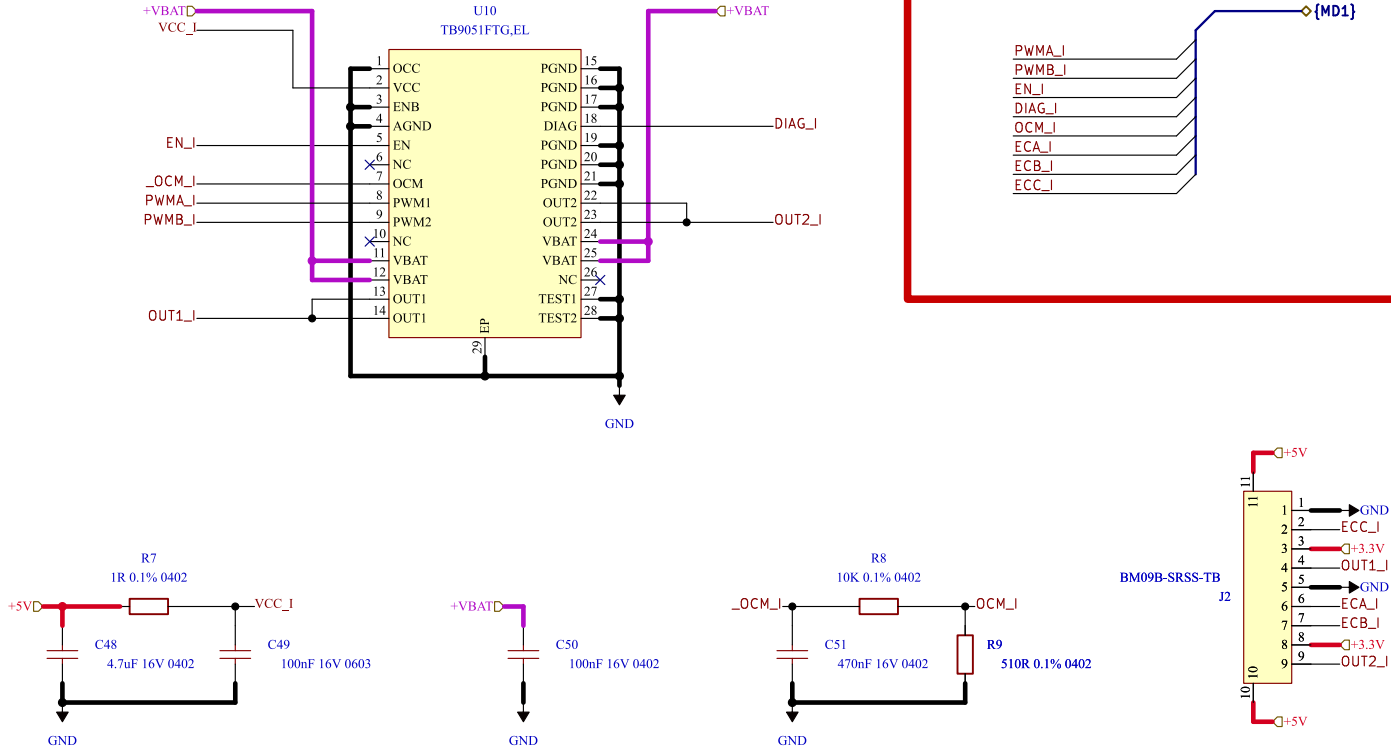
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Sheet Path:	Reviewer:		Size: A3	Sheet: 6 of 23	
/Project Architecture/Power - Generation/					

[7] Power - Connectors



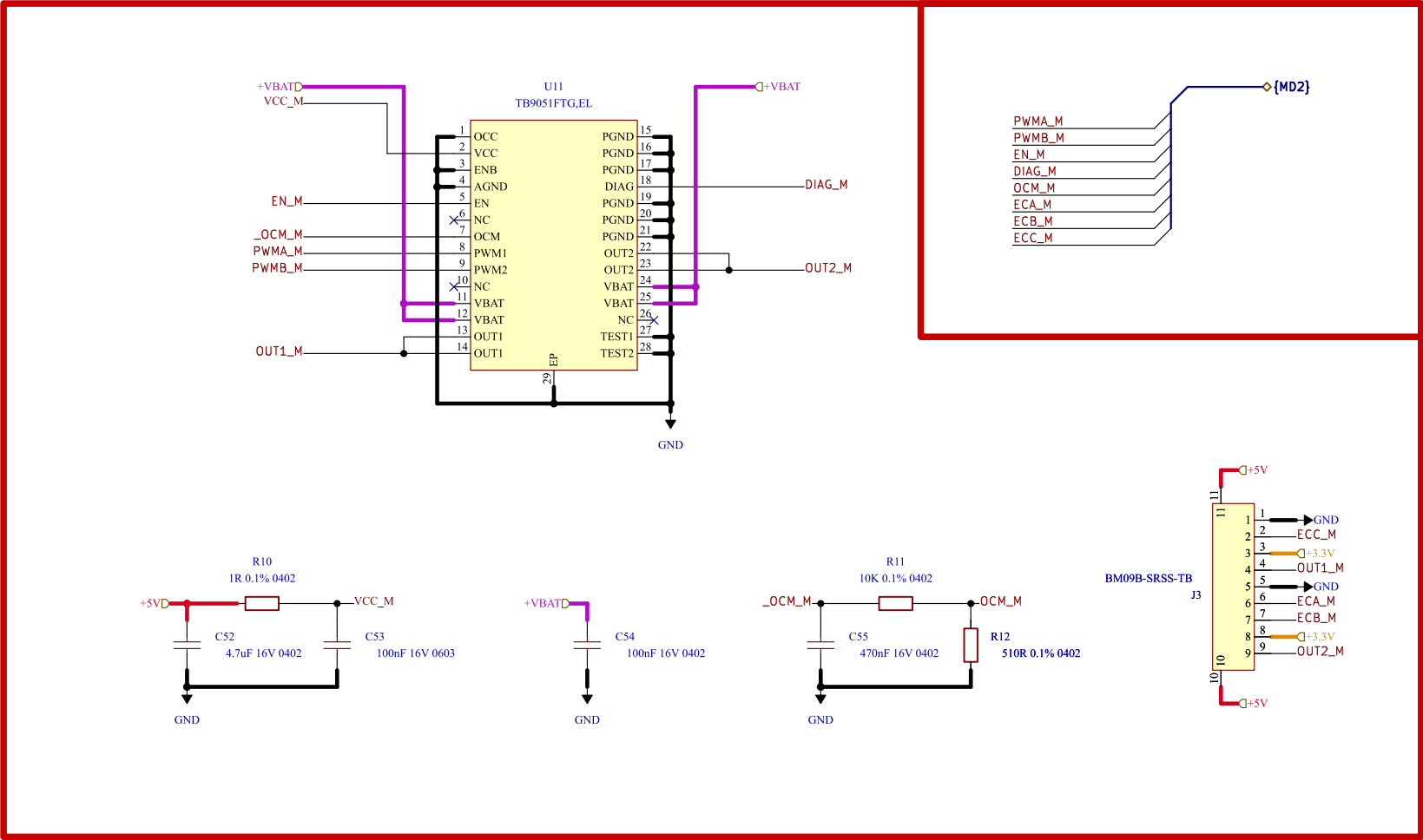
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Sheet Title:	File Name: Power - Connectors.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
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[8] Motor Control - Index



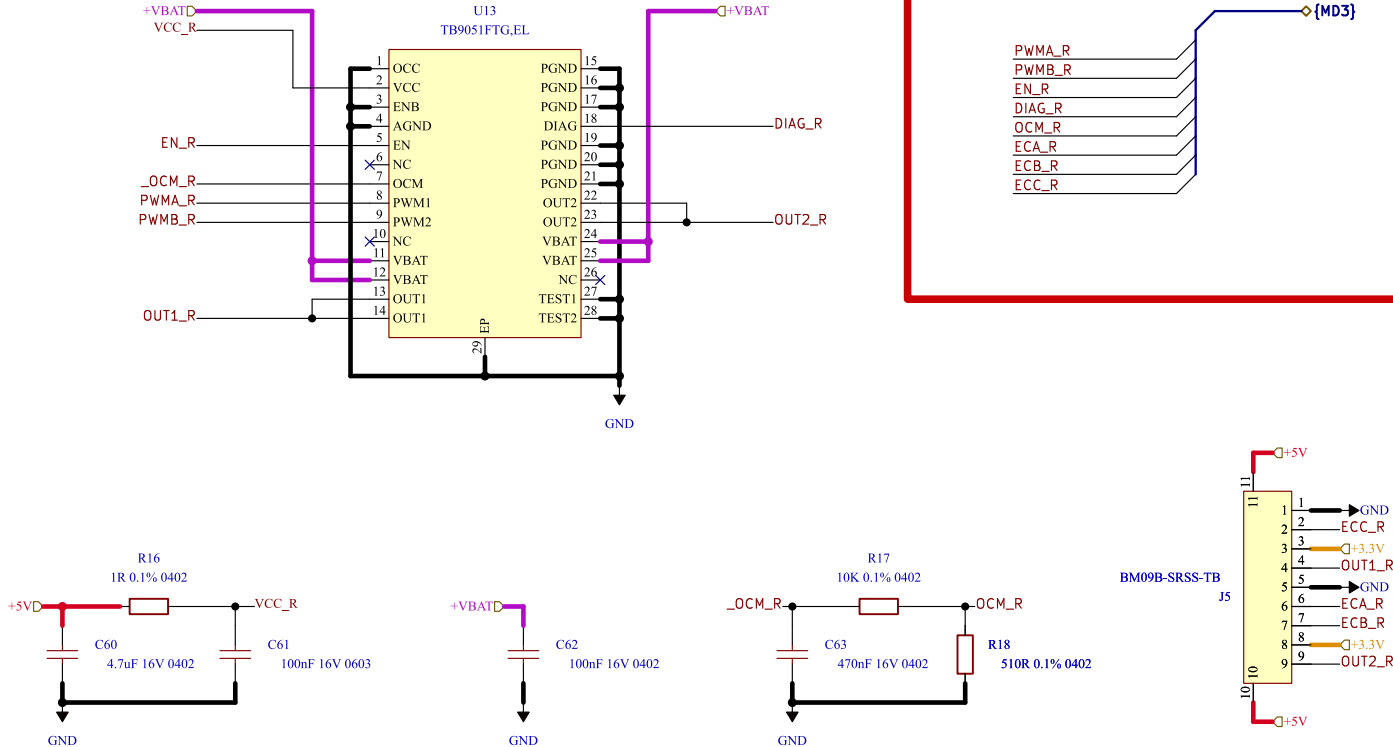
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Sheet Title:	File Name: Motor Control - Index.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
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[9] Motor Control - Middle



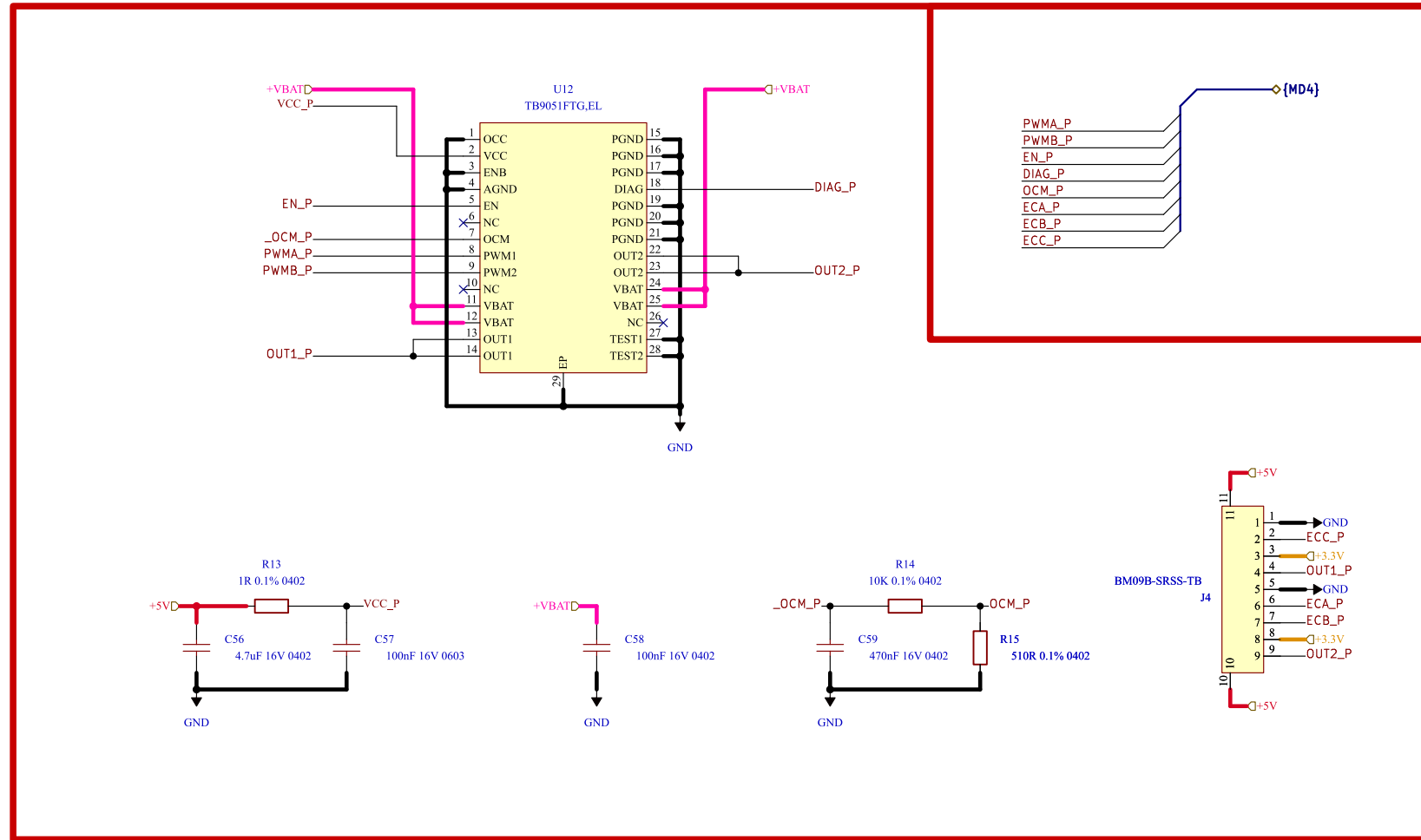
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[10] Motor Control - Ring



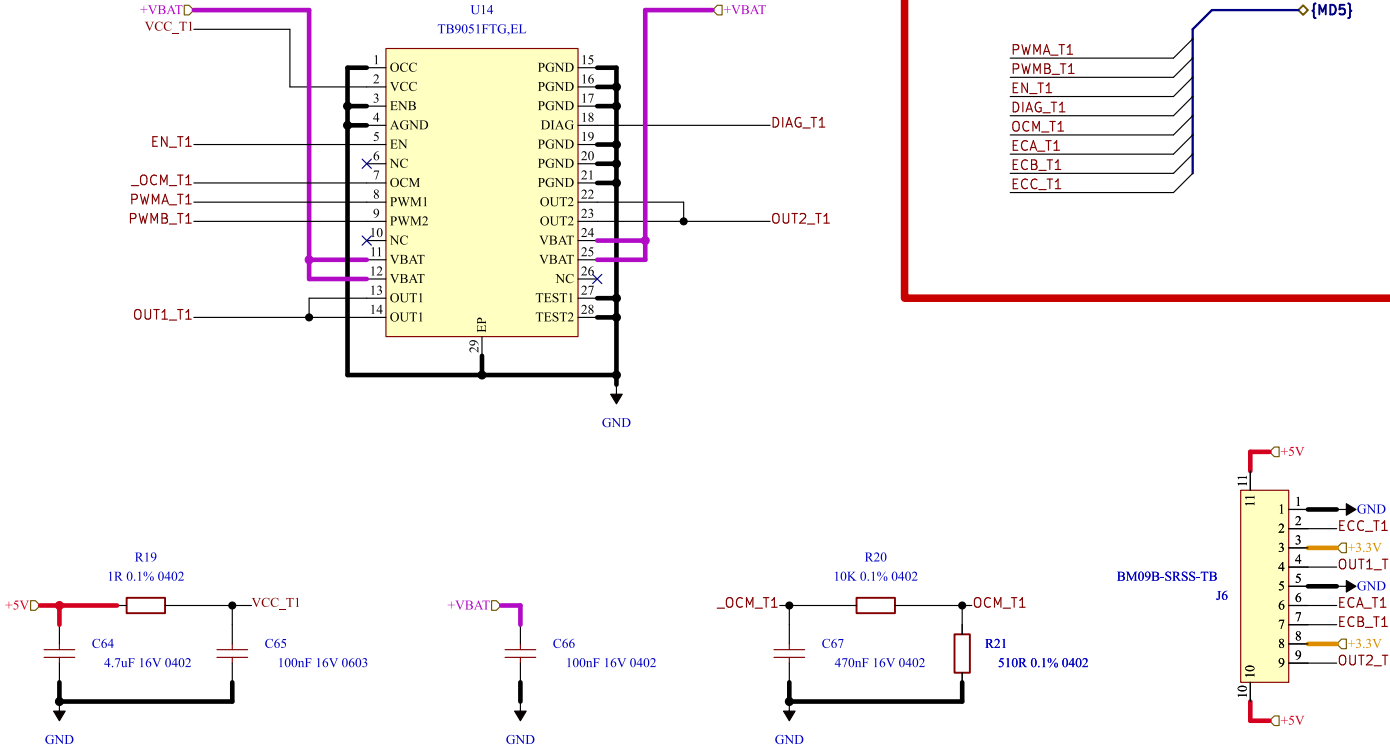
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Sheet Title:	File Name: Motor Control - Ring.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
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[11] Motor Control - Pinky



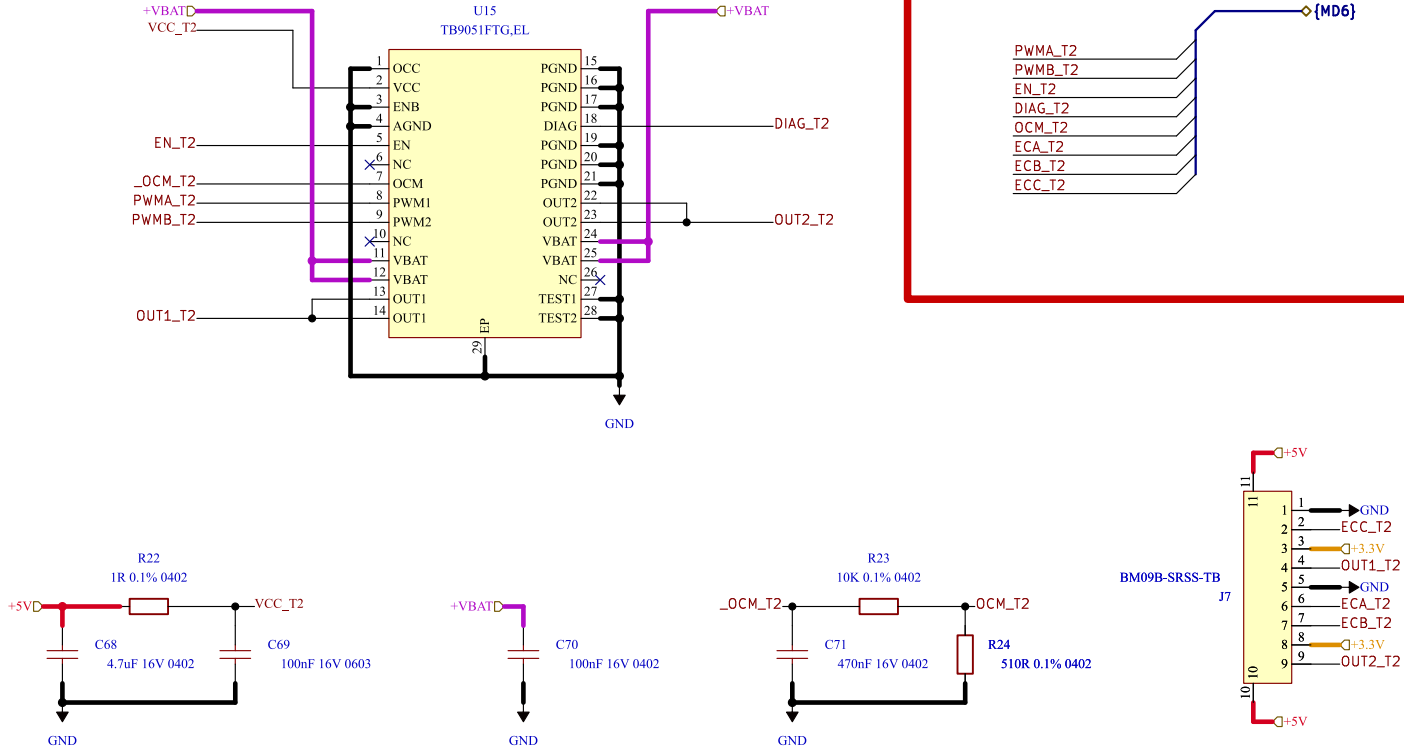
Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fe
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Sheet Title:	File Name: Motor Control - Pinky.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
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[12] Motor Control - Thumb 1



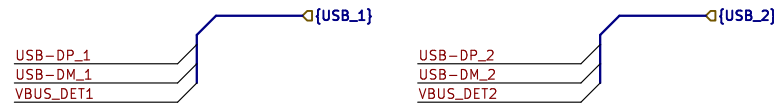
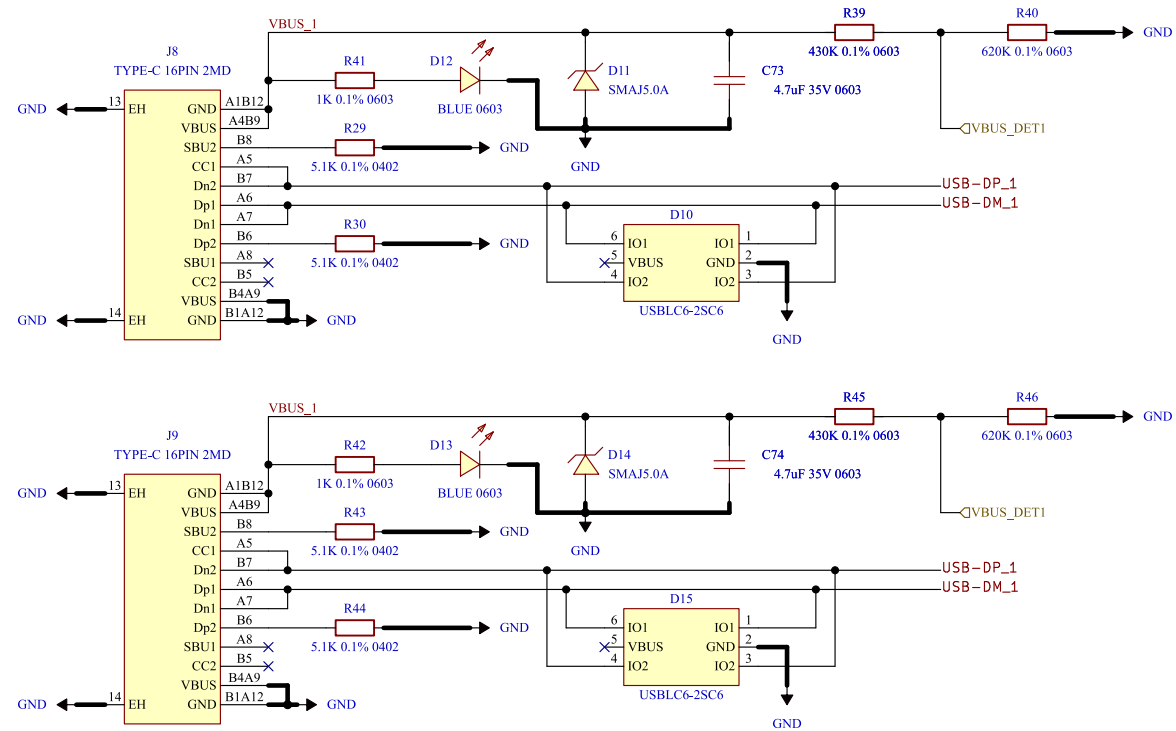
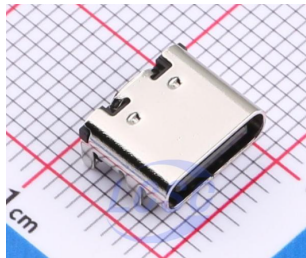
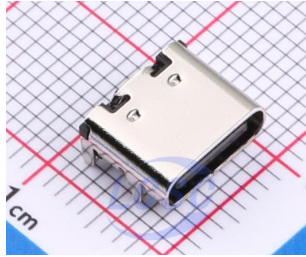
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	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Thumb 1.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Motor Control - Thumb 1/		Reviewer:	Size: A4	Sheet: 12 of 23		

[13] Motor Control - Thumb 2



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Thumb 2.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Thumb 2/		Reviewer:	Size: A4	Sheet: 13 of 23

[14] Interface - USB 2.0



DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 §7.2.3 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPU1 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

c99a2fe

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+ (Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

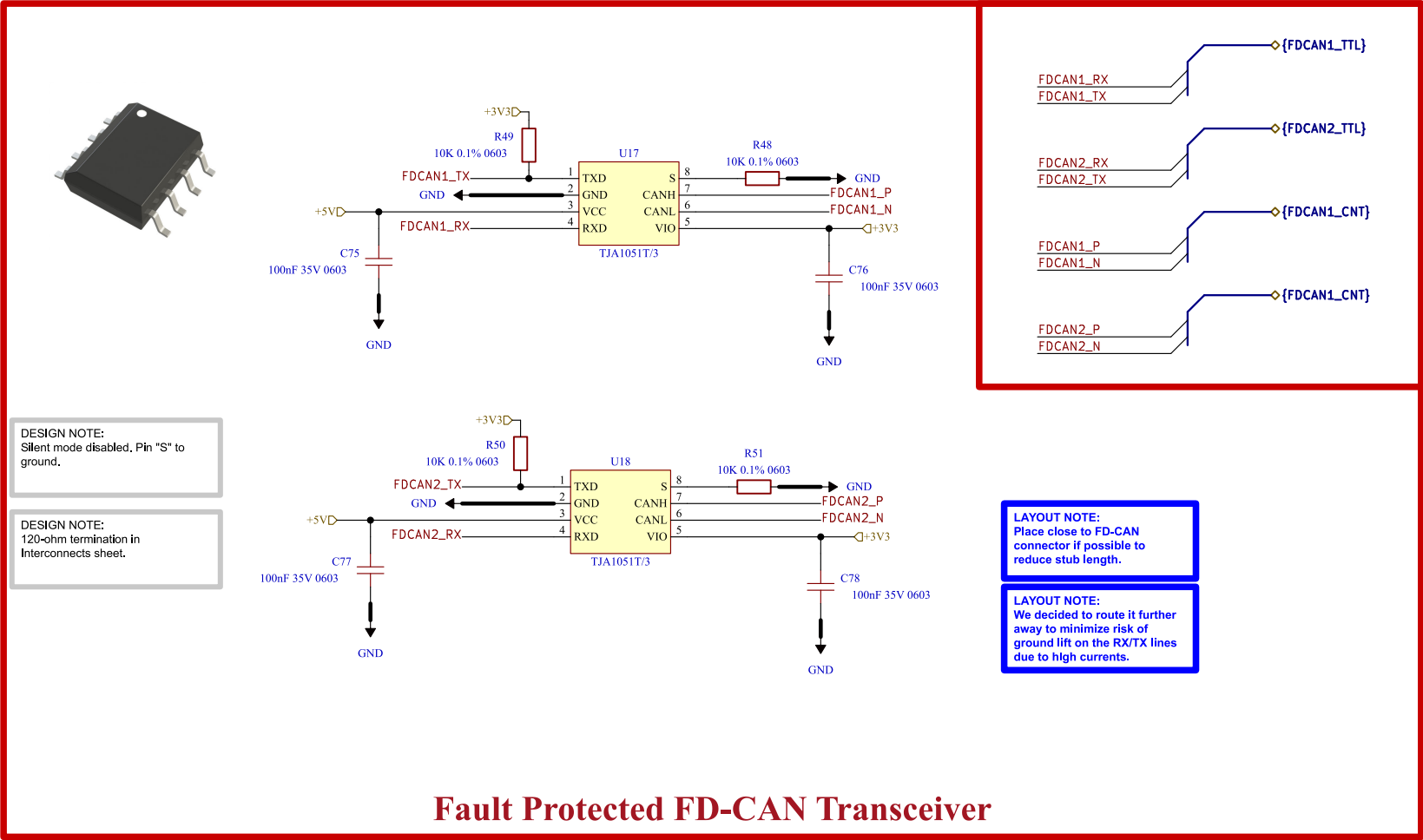
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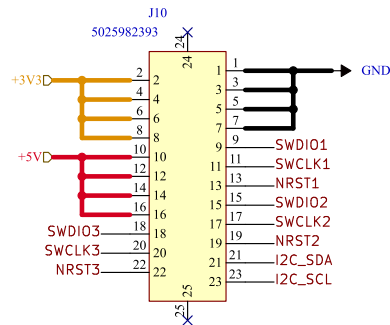
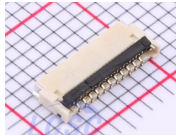
14 of 23

[15] Interface - FD - CAN

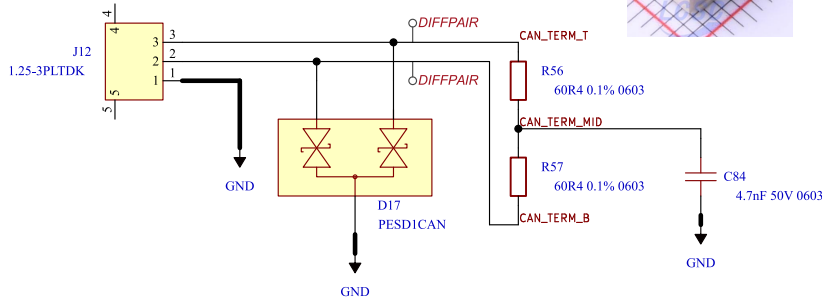


Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Interface - FD-CAN.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Interface - FD - CAN/		Reviewer:	Size: A4	Sheet: 15 of 23	

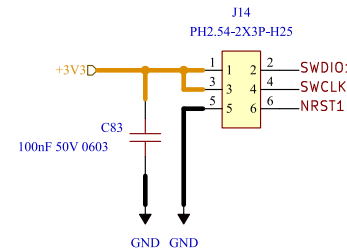
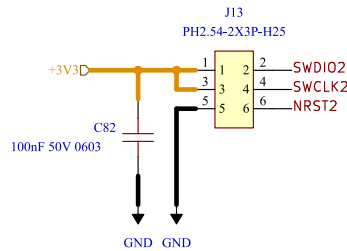
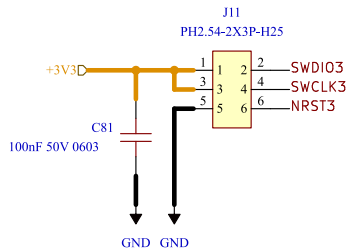
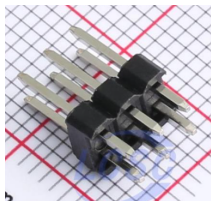
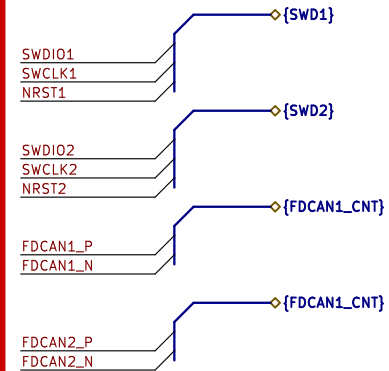
[16] Interface - Interconnects



FFC Connector MCB & MB



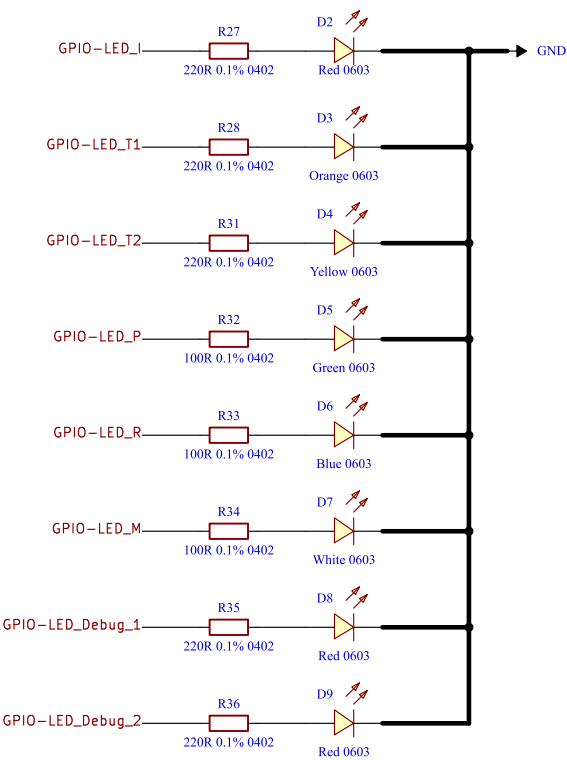
FD-CAN Connector and Termination



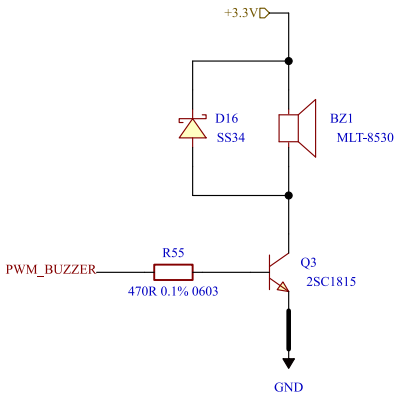
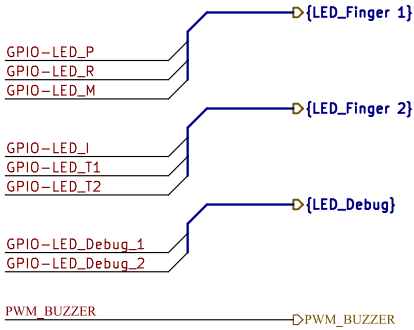
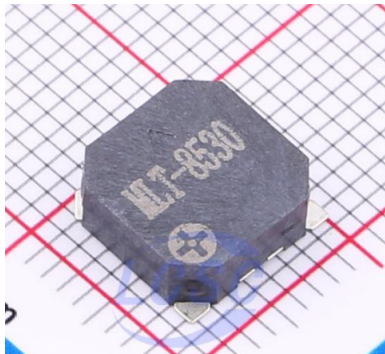
SWD Debug Connector

Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Interface - Interconnects.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A4	Sheet: 16 of 23

[17] User - LED Indicators & Buzzer



LED Color	Typical Vf Range
Red	1.8 to 2.1
Amber	2 to 2.2
Orange	1.9 to 2.2
Yellow	1.9 to 2.2
Green	2 to 3.1
Blue	3 to 3.7
White	3 to 3.4



Comments:

Company:

Willas



威倫電子股份有限公司
Willas Electronic Corp.

Variant:

DRAFT

Git Hash:

e99a2fe

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

User - LED Indicators & Buzzer.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+ (Unreleased)

Sheet Path:

/Project Architecture/User - LED Indicators & Buzzer/

Reviewer:

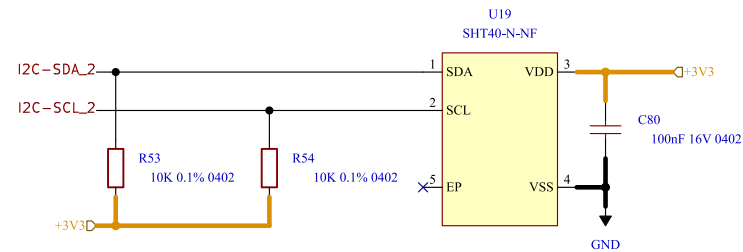
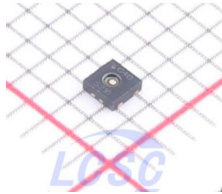
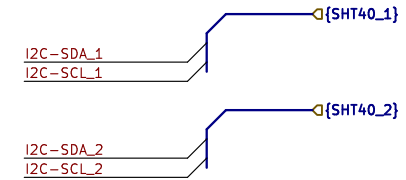
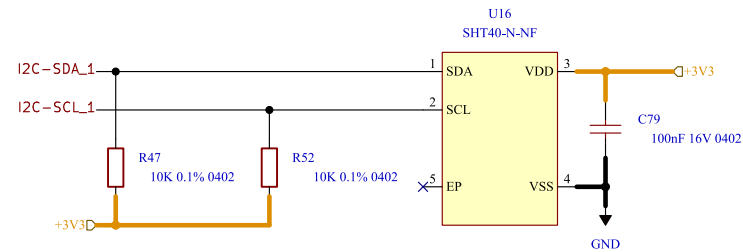
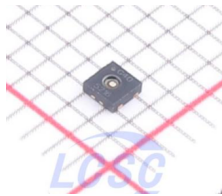
Size:

A4

Sheet:

17 of 23

[18] Sensing - Temperature & Humidity



Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: c99a2fe	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Sensing - Temperature.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Sensing - Temperature & Humidity/		Reviewer:	Size: A4	Sheet: 18 of 23		

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

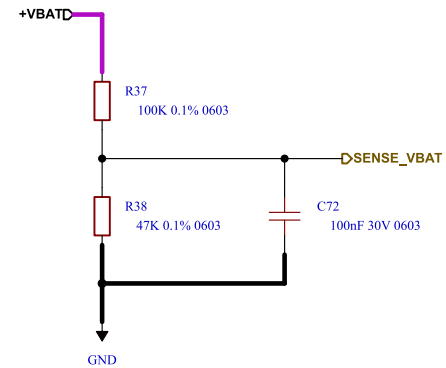
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

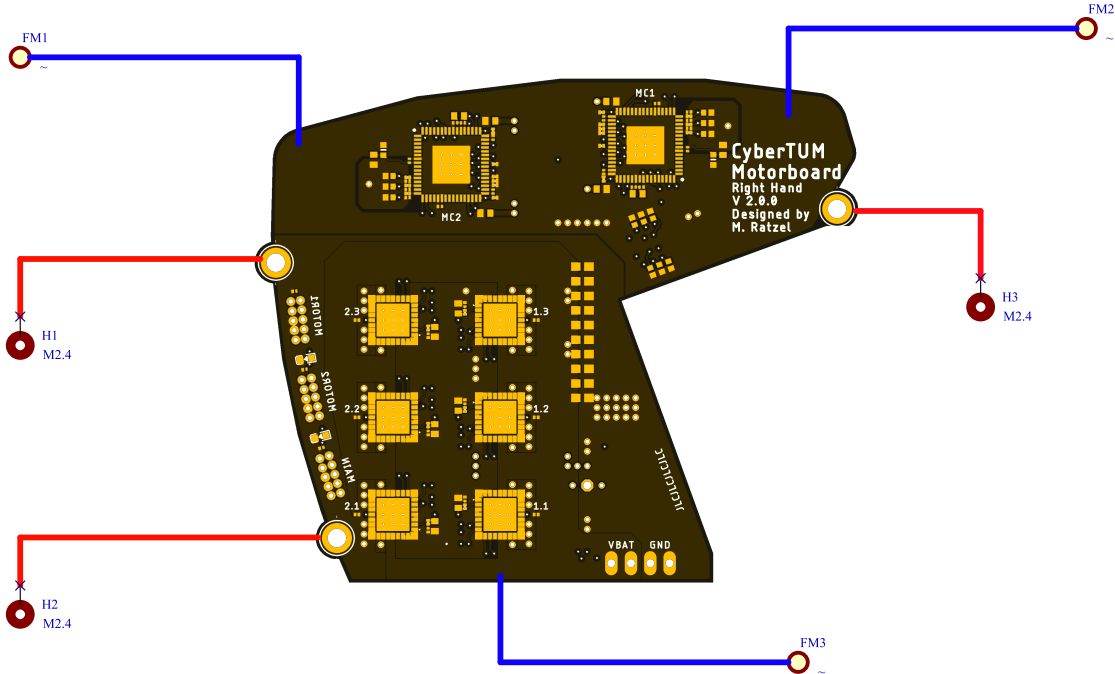
Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors ($\leq 100 \text{ ppm}/^\circ\text{C}$) for improved measurement accuracy.



Battery Voltage Sensing

Comments:	Company: Willas 		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Battery.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 19 of 23

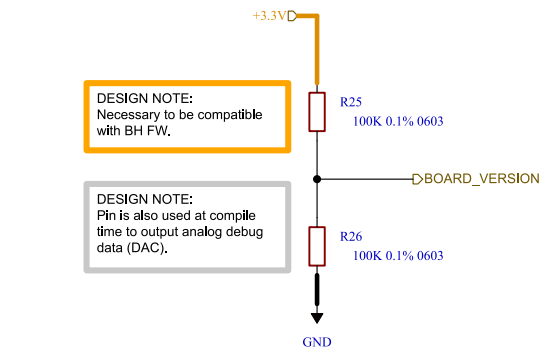
[20] Misc - Holes, Fiducials



Mounting Holes and Fiducial Markers

Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 20 of 23

[21] Misc - Board Version, DAC



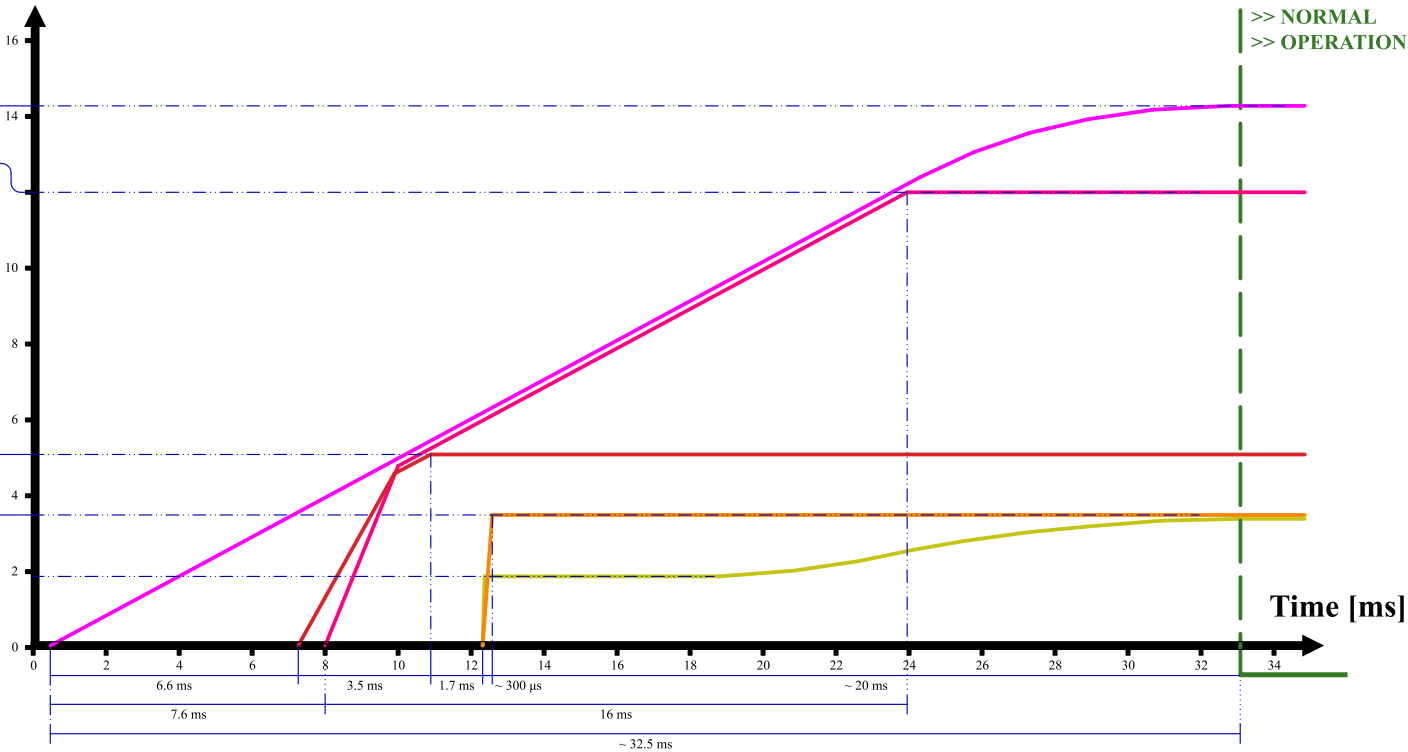
		TOP	BOTTOM
MCB	r1.1, r1.2	×	100k
	r1.3	100k	×
MB	v1.0-1.1	100k	100k

Board Version Resistor Divider

Comments:	Company: 威倫電子股份有限公司 Willas 		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Misc - Board Version DAC.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 21 of 23

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]



Comments:	Company: Willas		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 22 of 23

Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.	Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Revision History.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Revision History/			Reviewer:	Size: A4	Sheet: 23 of 23